

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

## Chapter 6: Simultaneous Linear Equations

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Solve  $x + y = 10$  and  $x - y = 4$ .
2. Solve  $2x + y = 8$  and  $3x - 2y = 5$  by substitution.
3. Solve  $5x + 2y = 16$  and  $3x - 4y = -6$  by elimination.
4. Solve  $4x - 3y = 5$  and  $6x = 5y + 7$ .
5. Solve  $\frac{3}{x} + \frac{2}{y} = 12$  and  $\frac{2}{x} + \frac{3}{y} = 13$ .
6. Solve  $37x + 43y = 123$  and  $43x + 37y = 117$ .
7. The sum of two numbers is 50 and their difference is 14. Find them.
8. A two-digit number is 4 times the sum of its digits. If 18 is added to it, the digits are reversed. Find the number.
9. 3 pens and 4 pencils cost ₹50, and 5 pens and 2 pencils cost ₹60. Find the cost of each.
10. A boat goes 24 km upstream in 6 hours and 28 km downstream in 4 hours. Find the speed of the boat in still water.

## Answer Key

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1.  $x = 7, y = 3$

2.  $x = 3, y = 2$

3.  $x = 2, y = 3$

4.  $x = 2, y = 1$

5.  $x = \frac{1}{2}, y = \frac{1}{3}$

6.  $x = 1, y = 2$

7. 32 and 18

8. 24

9. Pen ₹10, pencil ₹5

10. 5.5 km/h (stream 1.5 km/h)

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